

FIPS Code. Federal Information Processing Standard codes. Each state in the United States is assigned a 2-digit code. For example, “01” refers to Alaska, “37” refers to North Carolina, and “56” refers to Wyoming.

Fixed Effect Pooling. Fixed effect pooling is one method to combine two or more distributions of health impact or economic value estimates into a single new distribution. Fixed effect pooling assumes that there is a single true underlying relationship between these component distributions, and that differences among estimated parameters are the result of sampling error. Weights for the pooling are generated via inverse variance weighting, thus giving more weight to the studies that exhibit lower variance and less weight to the input distributions with higher variance. See Random Effects Pooling below for additional information regarding pooling techniques.

Fixed Radius. An option to interpolate air quality data points that uses all monitors within a fixed radius (or distance) of a given point of interest. All monitors are used and weighted by their relative distance.

Geographic Area. Designation of a grid definition for linkage to a specific health impact function.

GIS. Geographic Information System. A GIS is a system of hardware and software used for storage, retrieval, mapping, and analysis of geographic data.

Global Burden of Disease. The World Health Organization global burden of disease (GBD) study measures burden of disease using the disability-adjusted-life-year (DALY). This time-based measure combines years of life lost due to premature mortality and years of life lost due to time lived in states of less than full health. The DALY metric was developed in the original GBD 1990 study to assess the burden of disease consistently across diseases, risk factors and regions.¹¹

Grid Cell. One of the many geographic, or spatial, components within a Grid Definition. These can be regularly or irregularly shaped.

Grid Definition. A BenMAP-CE Grid Definition provides a method of breaking a geographic region into areas of interest (Grid Cells) in conducting an analysis. This can be done in two ways - by loading a Shapefile (a particular type of GIS file) or by specifying a regularly shaped grid pattern. These are referred to as Shapefile Grid Definitions and Regular Grid Definitions, respectively. Typically, a Shapefile Grid Definition is used when the areas of interest are political boundaries with irregularly shaped borders, while a Regular Grid Definition is used when the areas of interest are uniformly shaped grids (e.g., rectangles). All grid definitions must have a unique (i.e., non-repeating) column and row index.

¹¹ For more information on the GBD, see: http://www.who.int/topics/global_burden_of_disease.

Health Impact Function. A health impact function calculates the change in adverse health effects associated with a change in exposure to air pollution. Based on a C-R function, a typical health impact function has inputs specifying the air quality metric and pollutant; the age and subgroup characteristics of the population affected; and the incidence rate of the adverse health effect.

Incidence. The total number of adverse health effects in a geographic region in a given unit time. In BenMAP-CE, this is the total number of adverse health effects avoided due to a change in air pollution levels.

Incidence Rate. The background rate of a health effect per person in a given geographic region. The average number of adverse health effects per person per unit of time, typically a day or a year. The incidence rate must be expressed at the same time scale as the remainder of the health impact function. For example, a health impact function quantifying day-to-day changes in premature death must specify a daily death rate.

Income Growth Adjustment. Adjusting certain valuation functions to reflect increases in real income over time. Generally, an increase in real income implies an increase in the willingness to pay (WTP).

Interpolation. The process of estimating the air quality level in an unmonitored area by using one or more nearby air quality monitors. BenMAP-CE uses two types of interpolation procedures: one is to simply choose the closest monitor, the other is to use a technique called Voronoi Neighbor Averaging. These interpolation methods are discussed in more detail in Appendix B.

Lat/Long. Latitude and longitude information to specify the geographic coordinates of a spatial location. The CMAQ model data are usually provided for each grid cell identified by the latitude and longitude of the grid cell's center point. Latitude identifies the north-to-south location of a point on the Earth. Longitude identifies the east to west location of a point on the Earth.

Layer. In GIS, a layer represents a logical separation of mapped data usually representing a theme, such as political boundaries, roads, ozone data, number of mortalities avoided, etc.

Layer Statistics. The summary statistics that correspond to the active layer in BenMAP-CE. For example, “mean”, “standard deviation” or “max” of PM_{2.5} air quality grid.

Metadata. Data that serves to provide context or additional information about other data. BenMAP-CE stores a minimum set of standardized metadata fields for imported data files (e.g., file name, file date, reference, import date, and description). For certain data types, additional metadata are recorded. For example, GIS metadata will include information about datum, geographic coordinate system, resolution, and units.

Micrograms per Cubic Meter ($\mu\text{g}/\text{m}^3$). The unit of measure for particulate matter in the NAAQS. This unit represents the mass of PM and other particle pollutants found in a cubic meter of air.

Model Data. Pollutant concentration data that are generated by running an air quality model such as CMAQ. This is different from “monitor data,” which are based upon observed concentrations.

Monetize. In the context of human health benefits assessment, this is the practice of expressing society’s preferences for avoiding certain health effects as an economic value (e.g., in U.S. dollars). In BenMAP-CE we estimate monetized benefits by using either Willingness to Pay or Cost of Illness valuation functions (see above and below).

Monitor Data. Pollutant concentration data that are based upon measurements from an air quality monitor. “Raw” monitor data usually refers to data that are taken directly from measurement networks, with no additional processing of the data. Monitor data are different from “model data,” which are based upon numerical predictions from an air quality model.

Monitoring. Actual measurements of air pollution concentrations. The U.S. EPA has monitoring data, as well as other information related to monitoring, available through its Air Quality System (AQS): <https://www.epa.gov/aqs>.

Monte Carlo Simulation. A technique used in BenMAP-CE to quantify the confidence intervals around mean incidence and economic value estimate by randomly sampling an uncertainty distribution around the effect coefficients or willingness to pay estimates.

Morbidity. A measure of being diseased or afflicted by an illness (generally non-fatal).

Mortality. A measure of the number of deaths in a given population.

National Ambient Air Quality Standards (NAAQS). The U.S. EPA establishes levels for pollutants that are considered harmful to public health and the environment. The Clean Air Act established two types of national air quality standards. Primary standards set limits to protect public health, including the health of “sensitive” populations such as asthmatics, children, and the elderly. Secondary standards set limits to protect public welfare, including protection against decreased visibility and against damage to animals, crops, vegetation, and buildings. The U.S. EPA has set NAAQS for six principal pollutants, which are called “criteria” pollutants: carbon monoxide, lead, nitrogen dioxide, ozone, particulate matter ($\text{PM}_{2.5}$, PM_{10}), and sulfur dioxide.

Odds Ratio. A quantitative measure reported in epidemiology studies of the relationship between exposure to air pollution and a health outcome. Odds Ratios must be converted to beta coefficients to be used in BenMAP-CE

Ordinality. In relation to air quality monitors, ordinality refers to the number of monitor values in the season that can exceed your standard. For example, if we had set the ordinality to four, then a monitor can have as many as three daily averages (assuming that we are using the daily average metric to define our standard) greater than your standard without violating the standard. In terms of rollback, if it has more than three daily averages in exceedance of the standard, then the rollback technique will be applied to that monitor.

Ozone (O₃). BenMAP-CE focuses on ground level or “bad” ozone, which is not emitted directly into the air, but is created by chemical reactions between oxides of nitrogen (NO_x) and volatile organic compounds (VOCs) in the presence of sunlight. Emissions from industrial facilities and electric utilities, motor vehicle exhaust, gasoline vapors, and chemical solvents are some of the major sources of NO_x and VOC. Breathing ozone can trigger a variety of health problems, particularly for children, the elderly, and people of all ages who have lung diseases such as asthma. Ground level ozone can also have harmful effects on sensitive vegetation and ecosystems.

Particulate Matter. Particulate matter, also known as particle pollution or PM, is a complex mixture of extremely small particles and liquid droplets. Particle pollution is made up of a number of components, including acids (such as nitrates and sulfates), organic chemicals, metals, and soil or dust particles. Once inhaled, these particles can affect the heart and lungs and cause serious health effects. Includes PM_{2.5} (particles less than 2.5 microns in aerodynamic diameter), PM₁₀ (particles less than 10 microns in aerodynamic diameter), and PM_{10-2.5} (particles between 2.5 and 10 microns in aerodynamic diameter).

Parts per Million (ppm). This unit represents the concentration of the pollutant in a million parts of air. Carbon monoxide is often measured in units of ppm.

Parts per Billion (ppb). This unit represents the concentration of the pollutant in a billion parts of air. Ozone concentrations in BenMAP-CE are reported in units of ppb.

POC (Parameter Occurrence Code). An identifier used by U.S. EPA to distinguish between multiple monitors at the same site that are measuring the same parameter. For criteria pollutants, multiple monitors may be collocated to check precision. For combining data at the site level, the POC identifies the primary monitor (most frequent sampling). (POC appears in BenMAP-CE’s advanced filtering options for monitor data.)

Pooling. The combining of different sets of data. BenMAP-CE has several pooling methods, including fixed effects, fixed/random effects, and subjective weighting. Appendix K discusses the pooling approaches available in BenMAP-CE.

Point Mode. When defining the configuration, you may choose to either estimate adverse health effects in point mode or using percentiles. The point mode simply means that BenMAP-CE will use the mean value of the coefficient in the health impact function.